

# **Mechatronic Skripsie Project**

## **Biweekly Progress Report**

for the project

### **Development of a High-Power Audio Amplifier with an Automated Thermal Management System**

|                               |                               |
|-------------------------------|-------------------------------|
| <b>Student Name:</b>          | Karl Johan Hofmeyr            |
| <b>Student Number:</b>        | 27150526                      |
| <b>Module / Project Code:</b> | 478                           |
| <b>Lecturer / Supervisor:</b> | Ms. Tessa Hall                |
| <b>Report Number:</b>         | 04                            |
| <b>Reporting Period:</b>      | 08 April 2026 – 19 April 2026 |
| <b>Date Submitted:</b>        | 20 April 2026                 |

## 1 Overall Project Status

|                        |                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Overall Status:</b> | <b>On Track</b>                                                                                                      |
| <b>Current Phase:</b>  | Calculations                                                                                                         |
| <b>Report Summary:</b> | This reporting period focused on the calculations and requirements of the input stage and VA stage of the amplifier. |

## 2 Objectives for the Reporting Period

The main objectives for this two-week period were:

1. Complete calculations for the input stage of the amplifier.

## 3 Work Completed

### 3.1 Literature Review and Background Study

- Reviewed amplifier topology.
- Read Texas Instruments paper about bridge tied loads.
- Researched thermal control methods for heat management.

### 3.2 Circuit Design Progress

Circuit development completed:

- Designed constant current source.
- Designed current mirror.
- Designed input stage, with temporary VA stage.
- Considered feedback topology and compensation methods.

### 3.3 Thermal Control Progress

- Possible NTC thermistor and mosfet solution.
- Smallest ESP32 and 12V 4-pin PWM controlled fan.
- Temperature sensor already complete.
- Need to design fan driver circuit.

## 4 Work Planned for the Next Reporting Period

The planned work for the next two weeks is as follows:

1. Complete and test VA circuit in LTSpice.
2. Design cooling fan driver.

3. Complete calculations for current source and mirror in input stage.

## 5 Matters Requiring Lecturer / Supervisor Input

List any issues on which you would like feedback:

- Am I allowed to use the LM3886 IC?
- In a BTL/Parallel BTL configuration, where does the signal split? At what stage?
- What is linearity? Why is it good? Possibly in reference to the linear operation region of a transistor?
- What are the capabilities of the power supplies?
- Where could I obtain spice models for components?

**Student Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_